Geometry-Aware JEPA for EEG

A controlled frozen-transfer study on TUAB

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Hack the World(s) – EEG Track · Team **Hello Worlds**

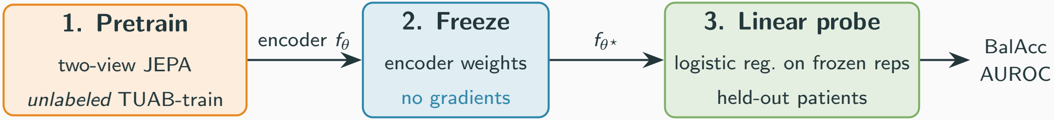
Motivation

The question we actually asked

- **EEG foundation models** are everywhere – but most report *fine-tuned* numbers. What survives when you **freeze** the encoder?
- **Self-supervised JEPA** avoids collapse with an *anti-collapse regulariser*. The EEG covariance lives on a curved **SPD manifold**, not in flat Euclidean space.
- **Our pre-registered hypothesis**: making anti-collapse *geometry-aware* (act in the SPD **tangent**) and *distribution-free* (**PEIRA**) beats a plain ambient baseline.
- **What we ship**: a controlled study with an *honest* answer – a strong frozen probe, a clean negative result with a geometric mechanism, and label/data-efficiency curves.

We are not a foundation model, not a world model, not SOTA in 24h.

The frozen-transfer protocol



- **Why frozen?** It measures the *representation*, not the fine-tuner. Apples-to-apples across methods, and the regime where many published FMs quietly collapse.
- **Selection rule**: hyper-parameters chosen on a patient-disjoint *train* dev-split; the 276-recording eval is touched **once**, config locked – no selection leakage.

Data

Data: TUH Abnormal EEG (TUAB)

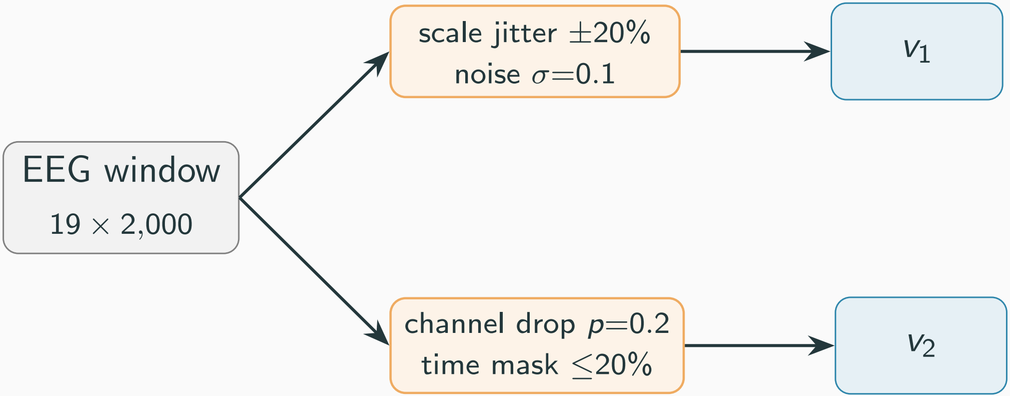
- **Task**: binary *normal* vs *abnormal* EEG, recording level.
- **Signal**: 19 channels @ 200 Hz; we sample 10 s windows (2,000 samples), per-channel z-scored.
- **Standard split**: 2,717 train / 276 eval recordings, **patient-disjoint** (2,076 vs 253 patients, overlap $= 0$ – verified).
- **Pretraining**: SSL reads only *unlabeled* train recordings; labels enter at the probe alone.

Why we don't chase accuracy

TUAB labels are noisy: inter-rater agreement $\sim$86–88%, no published Bayes ceiling. **Sub-1-point gaps without multi-seed CIs are not meaningful** – so we report *balanced accuracy on the full split*, 3 seeds, with error bars.

Two views for a JEPA invariance objective

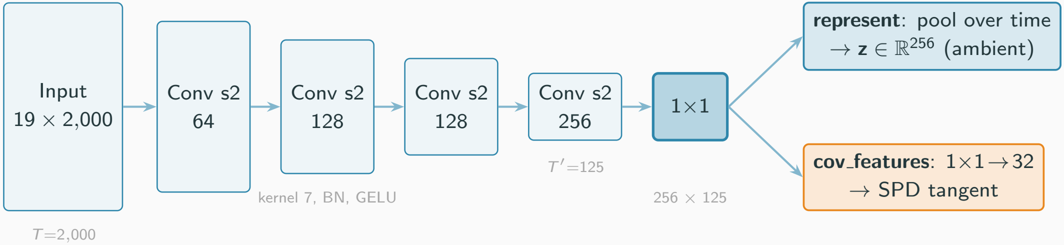
Each SSL sample = one 10 s window, augmented **twice** independently into views v_1, v_2:



- Augmentations (per channel, in z-scored units): amplitude scale jitter, additive

Architecture

Encoder: a 1D convolutional EEG backbone



4 strided `Conv1d` blocks (kernel 7, stride 2): $T : 2{,}000 \to 125$, width $19 \to 256$.

One backbone, two readouts – the probe always reads the pooled `represent`.

Where can anti-collapse act? Ambient vs SPD-tangent

Ambient (Euclidean).

Pool the feature map over time:

$$\mathbf{z} = \frac{1}{T'} \sum_t f_\theta(x)_{:,t} \in \mathbb{R}^{256}.$$

Regularise directly in $\mathbb{R}^{256}$. This is the **Laya-like** baseline.

Tangent (SPD log-Euclidean).

Per-window temporal covariance of a 32-dim projection:

$$C = \frac{1}{T'-1} \tilde{F} \tilde{F}^\top \in \mathrm{SPD}(32),$$

$$\mathbf{t} = \mathrm{vec}_{\sqrt{2}}\big(\log_m C\big) \in \mathbb{R}^{32(32+1)/2}.$$

Regularise in the tangent at the identity.

The clean knob

The probe reads the *same* pooled representation in both cases. Only the *space where anti-collapse is enforced* changes – this isolates the effect of **geometry**, nothing else.

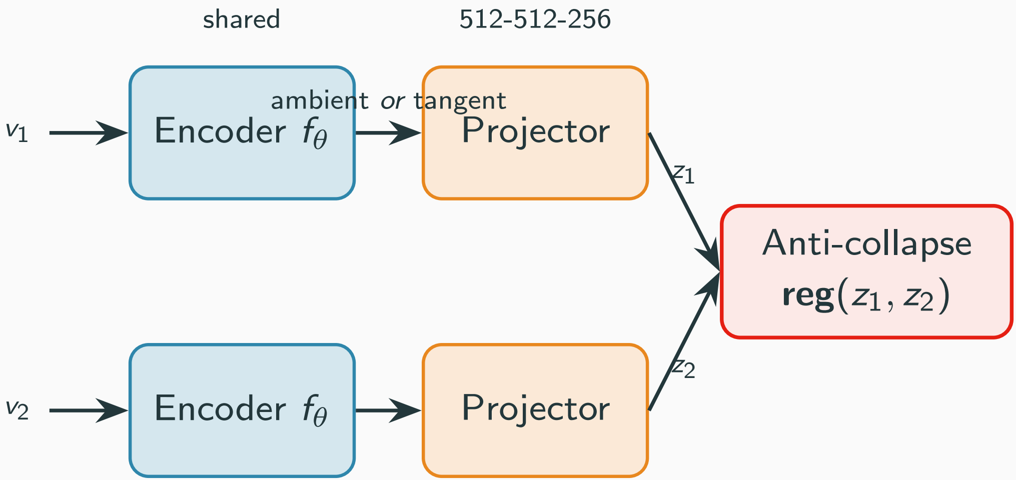
The ladder – a 2×2 (+ reference) factorial

regulariser \ **space**	**ambient** (Euclidean)	**tangent** (SPD)
VICReg (variance/cov)	`C0` reference	—
SIGReg (Gaussianity slices)	`C1` *Laya-like baseline*	`C2` geometry-aware
PEIRA (distribution-free)	`C3`	`C4` *ex-hypothesis*

- **SIGReg** (LeJEPA/BCS): tests random 1-D slices for Gaussianity – assumes an *isotropic* target.
- **PEIRA**: distribution-free anti-collapse – no target distribution to mis-specify.
- **Hypothesis**: the geometry-aware / distribution-free cells (`C2`, `C4`) beat the ambient baseline `C1`.

Training

Two-view JEPA pretraining



- AdamW, lr 10^{-3}, weight decay 10^{-6}
- Collapse monitored *every epoch*: effective rank, per-dim std, off-diagonal covariance

Collapse is the #1 risk – so we watched it

A degenerate encoder maps every window to (almost) the same vector. We log three diagnostics on each batch:

- **Effective rank** – entropy of the singular-value spectrum; $\to 1$ means collapse.
- **Per-dim std** – $\to 0$ means collapse.
- **Off-diagonal covariance** – redundancy across dimensions.

Rule

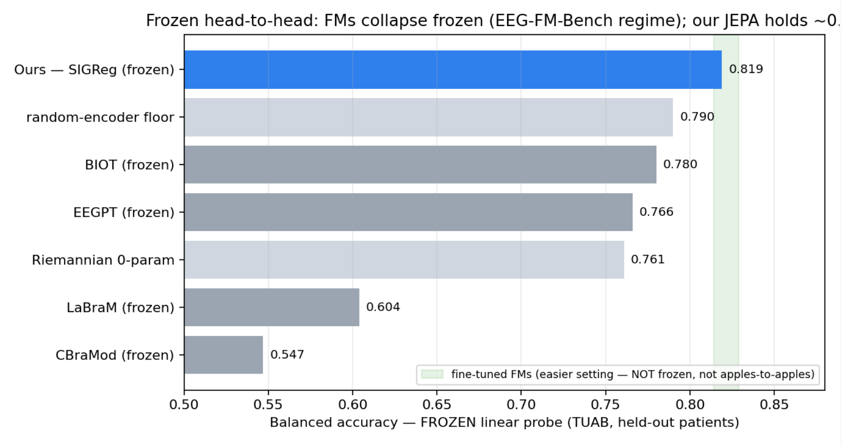
Kill flat runs early. The fix for collapse is *more* regularisation, not less. Every reported cell was checked non-degenerate (high eff. rank; PEIRA's tr_P rising).

Jury asset

"We didn't just avoid collapse – we *visualised* it."

Results

Frozen & in-domain: we hold where frozen FMs fall



Frozen linear probe, TUAB held-out patients. **Ours 0.819** (3-seed) sits above the random floor and every frozen FM baseline; fine-tuned FMs (green band) are an *easier*, non-comparable setting.

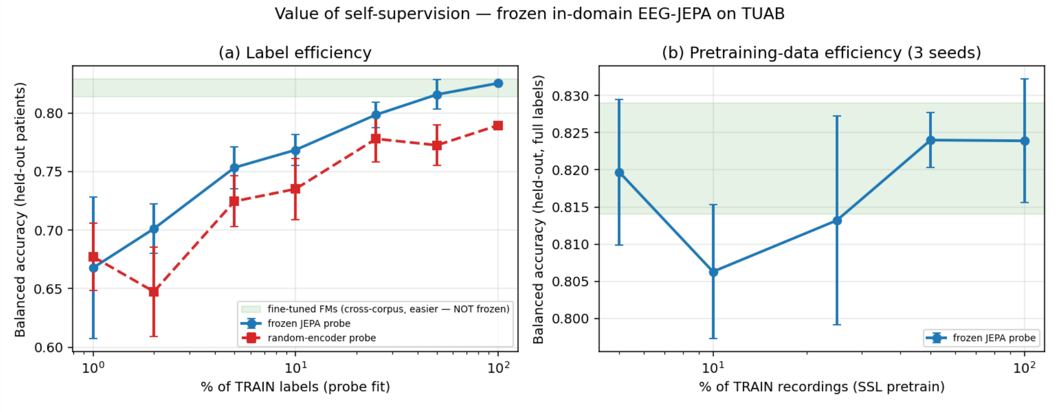
Apples-to-apples: it is not an in-domain artifact

- *"You pretrained on TUAB, the FMs didn't"* – the obvious attack on the in-domain number. So we ran the fair version.
- Pretrain SIGReg on a **general TUH corpus** (**TUSZ** seizure; TUAB-eval patients *excluded* $\Rightarrow$ patient-disjoint), freeze, linear-probe TUAB – the FMs' own regime (general pretrain $\rightarrow$ frozen TUAB).
- **Result: 0.814 BA / 0.889 AUROC** – essentially equal to in-domain 0.819. **The in-domain advantage is not the driver.**

Honest caveat

TUSZ is general *same-site* TUH (not multi-site like the FMs); single seed. We say "match while frozen", never "beat".

The value of self-supervision (label & data efficiency)



(a) The money plot: *JEPA @ 25% labels* $\approx$ *random-encoder @ 100%* – $\sim 4\times$ label efficiency, $\sim +0.04$ BalAcc. **(b)** Probe is stable down to small pretraining-data fractions.

The honest denominator: a random encoder already scores 0.79

- In our pipeline, a **random (untrained) conv encoder** + linear probe reaches $\sim$0.79 BalAcc.
- EEG abnormality is *power-driven*; random conv filters preserve band power. The strong frozen numbers are largely *architecture* + *task*.
- This floor is $\approx$ LaBraM-linear (0.795) and EEGPT-frozen (0.798), and *above* EEG2Rep / BIOT frozen (indicative – preprocessing differs).

Why disclose it?

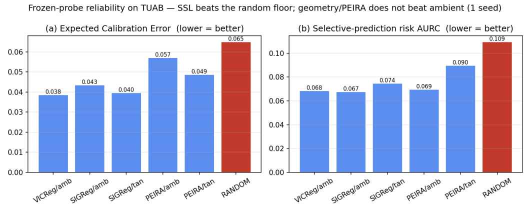
It pre-empts "is the SSL doing anything?", reframes our contribution as a **quantified** SSL increment ($\sim +0.04$ over random, $\sim 4\times$ label efficiency) – and is itself a sharp benchmark insight: *TUAB frozen-probe is easy for simple power features – in our consistent linear-probe setup, several published "foundation-model" frozen numbers barely clear this random baseline.*

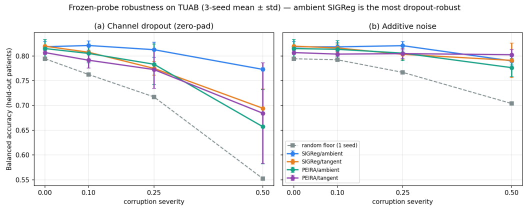
Where 0.819 sits – TUAB standard 2717/276 split

Setting	Protocol	BA	AUROC
Ours	**in-domain SSL, frozen, linear probe**	**0.819**	$\sim$0.90
LaBraM-Base / -Huge	fine-tuned	0.814 / 0.826	0.902 / 0.91
CBraMod	fine-tuned	0.829	0.923
REVE-Base	fine-tuned	0.832	0.925
LaBraM-Base	linear probe	0.795	0.884
REVE	linear probe	0.810–0.821	—
EEGPT	frozen / linear-probe-style	0.798	0.872
EEG2Rep	linear probe	0.766 (acc)	0.832
BIOT	linear probe	0.751 (acc)	0.829
classical Riemannian (Gemein 2020)	cov + tangent + classifier	$\sim$0.86 (acc)	—

Top of the frozen linear-probe results; lower edge of the *fine-tuned* BA band. AUROC $\sim$0.90 above BIOT/EEGPT, below CBraMod/REVE. **Competitive at the operating point – not SOTA.**

Beyond accuracy: calibration & robustness (3 axes, not 1)





- **Calibration**: every SSL encoder beats the random floor (ECE, selective prediction) – geometry/PEIRA does *not* beat ambient.
- **Robustness (3-seed)**: ambient SIGReg is the *most* channel-dropout-robust; the SPD-tangent objective *degrades* dropout-robustness; no geometry win under noise (the single-seed hint did not replicate).
- **The null holds on all three axes** – accuracy, calibration, robustness. Geometry buys none of them here.

Insight

The result we pre-registered against: a clean null

3-seed mean balanced accuracy, $\{1, 1000, 10000\}$:

reg \ space	**ambient**	**tangent**
VICReg	0.814	—
SIGReg	**0.819** (locked)	0.820†
PEIRA	0.815	0.807

- Every cell $\approx 0.82\pm$ seed noise; per-cell ranges overlap heavily. **No effect of regulariser, no effect of space, no interaction.**
- Our hypothesis (*geometry / PEIRA win*) is **falsified**. We *locked* ambient SIGReg and kept the 2×2 as the published ablation.

† caveat: SIGReg's slice schedule is seeded per-step, not per-run, so its "tightest variance" is partly an artifact – we do not lean on it (see appendix).

Why the naive tangent is mis-specified (the mechanism)

On the SPD manifold, the natural Gaussian is a **wrapped Gaussian** – *non-isotropic* by construction (de Surrel et al. 2025, Def. 4.1: $t \sim \mathcal{N}(\mu, \Sigma)$ with full Σ, not $\sigma^2 I$).

SIGReg forces *isotropy* ($\mathcal{N}(0, I)$) where the geometry induces a non-isotropic covariance – it fights the structure that carries the signal.

	ours	**principled**
target	isotropic	wrapped, full Σ
base pt	identity	Fréchet mean

Honest framing

Future-work *motivation*, not a validated mechanism: at $n{=}3$ the tangent effect is null, so our data *cannot* confirm the wrapped-Gaussian prediction.

And the principled object is *already* our Riemannian baseline ($0.761 \ll 0.82$): the linear probe re-derives second-order tangent structure for free, so best case for tangent SSL is a *tie*.

We measured the geometry – not just hid it

- The probe reads the first-order `represent`; the SPD-tangent lives *only* in the SSL loss. So we exposed the **learned tangent** to the probe directly.
- Learned-tangent probe: **0.79** BA (SIGReg-tangent 0.790 $>$ its random-tangent floor 0.776) – a *real but weak* second-order signal, **below** the first-order rep (0.82).
- TUAB rewards first-order power; the second-order tangent caps $\sim$0.78–0.80 (cf. MENDR 0.78–0.80, Riemann 0.761). **The tangent is not broken – it is capped by the task.**

What we claim – and what we don't

DO claim

- Strongest among published *frozen*/linear-probe TUAB baselines; inside the fine-tuned BA band.
- A clean **3-seed null** on geometry/PEIRA, with a geometric mechanism + precedent.
- In-domain SSL is the (disclosed) reason it is strong: $\sim +0.04$ over random, $\sim 4\times$ label efficiency.

DON'T claim

- ~~SOTA on TUAB~~ – AUROC below best; preprocessing differs.
- ~~We beat CBraMod/LaBraM~~ – they are fine-tuned; say "*match while frozen*".
- ~~Tangent-SPD is useless for EEG~~ – only "no benefit in our TUAB frozen pilot".
- ~~TUAB is saturated at X~~ – ceiling unknown.

The boxed claim

Domain-matched JEPA pretraining yields a *frozen* encoder whose linear probe (0.819 BA / $\sim$0.90 AUROC, 3 seeds) is competitive with the TUAB *fine-tuning* literature and stronger than typical *frozen*-probe baselines. In a controlled 2×2 ablation, neither PEIRA nor a geometry-aware tangent improved over ambient SIGReg.

Limitations & Future Work

Limitations & Future Work

- **Single dataset.** Specialization $\neq$ generality. The honest generality test is frozen cross-dataset transfer (TUEV/Sleep-EDF) – started, but left as future work rather than over-claimed.
- **Underpowered null.** $n=3$ seeds; the inter-cell gaps (≤ 0.013) are below the eval's bootstrap sampling noise ($\pm$0.02–0.03). The honest statement is "*benchmark-resolution-limited*", not "geometry is dead".
- **The principled fix.** A wrapped-Gaussian SIGReg on the *channel* covariance, AIRM tangent at the running Fréchet mean – $\sim$40 LOC + one re-pretrain, with real `eigh`-backward numerical risk.
- **Where geometry should pay off.** Robustness, calibration, and transfer under montage/threshold shift – *not* a balanced-accuracy race.

Credit & honest scope

Credit & honest scope: what 24h bought (and what it didn't)

Built on `eb_jepa` – and it carries the baseline.

- Our encoder, SSL loop, and the *clean EEG data pipeline* are `eb_jepa` (the framework we were handed). Our work is an *additive* comparative study on top – geometry/PEIRA cells, reliability axes, latent viz – *not* a reimplementation.
- **A large share of the frozen-probe strength is `eb_jepa`'s**, not ours: its preprocessing + JEPA implementation do much of the heavy lifting (it makes TUAB tractable under a *frozen* probe). We claim the *study*, not the raw accuracy.

We do not claim SOTA.

- 24h = a frozen linear-probe study on *one* task (TUAB), TUAB-tuned pipeline. Cross-model numbers are **indicative, not controlled re-runs** of others' pipelines.
- Laya (recent LeJEPA SOTA): **29,109 h / 20,940 subjects / 17 montages / 14 tasks** under the *same* frozen contract – a different league of *generality*. Our single-task value $\neq$ their generalist value.
- **What 24h bought**: a clean multi-axis *negative*, *healthy-training* evidence it is not a broken run, and a *manifold-aware* latent viz (AIRM Riemannian t-SNE) – a methods contribution, not a leaderboard claim.

Conclusion

Conclusion

- A **controlled frozen-transfer study**, not a leaderboard entry.
- **A strong frozen probe** (0.819 BA, $\sim$0.90 AUROC) that holds where published FMs collapse when frozen – with the random-encoder floor honestly disclosed.
- **A clean, pre-registered negative result**: *where* anti-collapse acts (ambient vs SPD-tangent) and *which* regulariser do not change frozen-probe accuracy on TUAB – with a geometric reason why.
- **The null is multi-axis**: geometry/PEIRA help neither accuracy, nor calibration, nor robustness (3 seeds) – and it holds even *general*-pretrained (TUSZ $\rightarrow$ frozen TUAB, 0.814).
- **Honesty as the headline**: an honest negative beats a fragile positive.

Built on a trimmed `eb_jepa` vendor. Team **Hello Worlds**.

References

References i

- Panchavati, S. et al. (2026). *Laya: a LeJEPA approach to EEG via latent prediction over reconstruction*. Preprint. *(our ambient baseline)*
- de Surrel, T., Lotte, F., Chevallier, S. & Yger, F. (2025). *Wrapped Gaussian on the manifold of Symmetric Positive Definite Matrices*. Preprint.
- de Surrel, T. *r_tSNE_in_C* – Riemannian t-SNE on SPD matrices (AIRM). github.com/thibaultdesurrel/r_tSNE_in_C. *(our manifold-aware latent viz)*
- Arbel, M., Terver, B. & Ponce, J. (2026). *PEIRA: Learning Predictive Encoders through Inter-View Regressor Alignment*. Preprint.
- *EEG-ReMinD: Riemannian SPD self-supervised EEG* (2025). Preprint. *(reconstruction, not JEPA)*

References ii

- LeCun, Y. (2022). *A Path Towards Autonomous Machine Intelligence (JEPA)*. OpenReview.
- Jiang, W.-B. et al. (2024). *LaBraM: Large Brain Model for EEG*. ICLR.
- Wang, J. et al. (2024). *CBraMod: Criss-Cross Brain Foundation Model*. Preprint.
- Yang, C. et al. (2023). *BIOT: Biosignal Transformer*. NeurIPS.
- Wang, G. et al. (2024). *EEGPT: Pretrained Transformer for Universal and Reliable Representation of EEG Signals*. NeurIPS.
- Foumani, N. et al. (2024). *EEG2Rep: Enhancing Self-supervised EEG Representation Through Informative Masked Inputs*. KDD.
- Gemein, L. et al. (2020). *Machine-learning-based diagnostics of EEG pathology* (Riemannian baseline). NeuroImage.

References iii

- Bardes, A., Ponce, J. & LeCun, Y. (2022). *VICReg*. ICLR.

Appendix

Appendix: encoder block + readouts

Each backbone block: `Conv1d`(kernel 7, stride 2) $\rightarrow$ `BatchNorm` $\rightarrow$ `GELU`. Widths $19 \rightarrow 64 \rightarrow 128 \rightarrow 128 \rightarrow 256$; a 1×1 head fixes the channel dim. Time $T : 2{,}000 \rightarrow 125$.

- `represent(x)` = time-pooled feature map $\in \mathbb{R}^{256}$ – the **only** thing the probe ever reads.
- `cov_features(x)` = a 1×1 projection to 32 channels, fed to the tangent map. $32 < 125$ keeps the covariance full-rank.
- Projector for SSL: `512-512-256` MLP (Linear + BN + GELU), discarded after pretraining.

Appendix: the SPD-tangent construction

From the feature map $F \in \mathbb{R}^{32 \times T'}$ (centred over time):

$$
\begin{aligned}
C &= \tfrac{1}{T'-1} FF^\top + \varepsilon I \in \mathrm{SPD}(32) && \text{(temporal covariance)} \\
\log_m C &= U \,\mathrm{diag}(\log \lambda_i)\, U^\top && \text{(via eigendecomposition)} \\
\mathbf{t} &= \mathrm{vec}_{\sqrt{2}}(\log_m C) \in \mathbb{R}^{32(32+1)/2} && \text{(off-diagonals scaled by } \sqrt{2}\text{)}
\end{aligned}
$$

- The $\sqrt{2}$ weighting makes the Euclidean norm of $\mathbf{t}$ equal the Frobenius norm of $\log_m C$ – the correct isometry for both LE and AIRM *at the identity*.
- At the identity base point, AIRM-log $\equiv$ Log-Euclidean: the metric is *not* an independent axis. Mis-specification = isotropic target + identity base point (two axes, not three).
- `eigh`-backward numerics: ε-clamped eigenvalues, small 32, float-safe; verified finite on GPU in the smoke test.

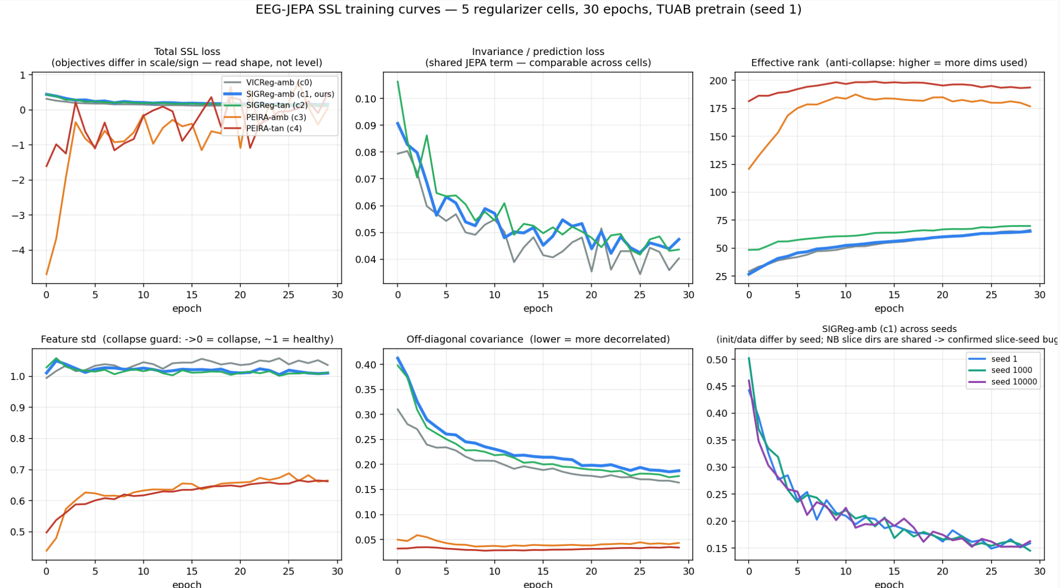
Appendix: collapse diagnostics

On a batch of embeddings $Z \in \mathbb{R}^{N\times D}$ (mean-centred), with singular values s_i and $p_i = s_i / \sum_j s_j$:

$$\text{eff. rank} = \exp\Big(-\sum_i p_i \log p_i\Big), \quad \text{feat. std} = \overline{\mathrm{std}(Z_{:,d})}, \quad \text{off-diag} = \frac{\|\,\mathrm{Cov}(Z)-\mathrm{diag}\,\|_F}{\sqrt{D(D-1)}}.$$

- Collapse $\Rightarrow$ eff. rank $\to 1$, feat. std $\to 0$. Healthy SIGReg-ambient run: eff. rank $27 \to 65$ over training.
- For PEIRA, the loss value is a *surrogate-gradient* number (can be negative, non-monotone). Only tr_P and eff. rank are valid progress signals.
- Per-cell non-degeneracy was audited before averaging – tangent-PEIRA (0.807) was confirmed healthy, not a partial collapse.

Appendix: SSL training curves (all cells healthy)



Appendix: evaluation protocol

- **Feature extraction**: 16 evenly-spaced 10 s windows per recording $\rightarrow$ pooled `represent` $\rightarrow$ averaged to a recording-level vector.
- **Probe**: `StandardScaler` + logistic regression on the frozen features. **Frozen encoder, no fine-tuning.**
- **Split**: train recordings fit the probe; the 276-recording eval (held-out patients) is scored **once**. Hyper-parameters selected on a patient-disjoint *train* dev-split only.
- **Metrics**: balanced accuracy (primary) + AUROC, on the full 2,717/276 split.
- **Floor**: an untrained random encoder, same probe, gives the $\sim$0.79 honest denominator.

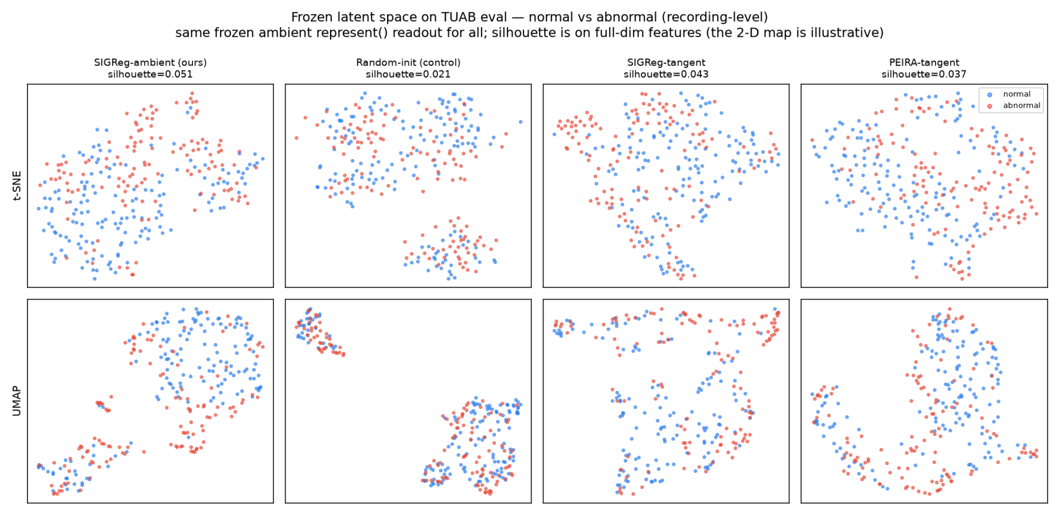
Appendix: label-efficiency numbers

Balanced accuracy vs. % of TRAIN labels used to fit the probe (frozen encoder fixed):

% labels	1	2	5	10	25	50	100
frozen JEPA	0.668	0.701	0.753	0.768	**0.799**	0.816	0.825
random encoder	0.677	0.647	0.725	0.735	0.778	0.773	**0.790**

- **JEPA @ 25% labels (0.799) $\approx$ random @ 100% (0.790)** – the $\sim 4\times$ label-efficiency claim.
- The SSL increment ($\sim +0.04$ at full labels) is small but consistent, and corroborated by a $\sim 2\times$ higher latent-space silhouette (appendix figure).

Appendix: frozen latent space (ours vs random vs geometry)



t-SNE and UMAP of frozen `represent` features on the eval set – same frozen ambient readout for all

Appendix: manifold-aware latent geometry (AIRM vs Euclidean)

Appendix: caveats we disclose to the jury

- **Report 0.819, not 0.833.** The single-seed 0.833 (SIGReg-ambient, seed 1) was the *top* of that cell's range; the 3-seed mean is 0.819.
- **SIGReg slice-seed bug (confirmed).** `BCS` seeds its random slice directions from `self.step`, not the run seed – so the slice schedule is identical across the 3 seeds. SIGReg's error bars don't include slicing randomness; its "tightest variance" is partly an artifact. The fix must precede any re-run (it shifts all SIGReg numbers).
- **PEIRA tangent (0.807) is not evidence against wrapped-Gaussian.** PEIRA is distribution-free $\Rightarrow$ immune to the isotropic-target mis-specification $\Rightarrow$ it never tests that hypothesis. 0.807 vs 0.815 is within noise.
- **Bootstrap CI.** Resampling the 276 eval recordings gives a ± 0.02–0.03 CI – wider than the max inter-cell gap (0.013). The null is benchmark-resolution-limited.
- **Preprocessing differs** from the published FM rows – the head-to-head is indicative, not a controlled re-run of their pipelines.